



Java

Building Robust Applications with Java Programming

Java is a powerful, versatile, and widely-used programming language known for its portability, performance, and extensive libraries. It is an object-oriented language that enables developers to build robust, scalable applications ranging from web services to mobile apps.

Java was created by James Gosling and his team at Sun Microsystems in 1995. It was designed to have as few implementation dependencies as possible, making it a “write once, run anywhere” language. This means that compiled Java code can run on all platforms that support Java without the need for recompilation.

Key Features of Java:

- **Object-Oriented:** Everything in Java is an object, which helps in modularizing and reusing code.
- **Platform-Independent:** Java applications are compiled into bytecode, which can run on any device equipped with the Java Virtual Machine (JVM).

- **Robust:** Java provides strong memory management and exception handling, reducing the chances of crashes.
- **Multithreaded:** Java supports multithreading, allowing concurrent execution of two or more threads for maximum utilization of CPU.
- **Security:** Java has several built-in security features, making it a secure language for networked applications.

Setting Up Java

Before you start coding in Java, you need to set up your development environment. Here’s how you can get started:

Installing Java Development Kit (JDK):

1. Download the JDK:

- Go to the Oracle JDK website (<https://www.oracle.com/java/technologies/javase-jdk11-downloads.html>).
- Download the latest JDK for your operating system.